



COLUMBIA | ENGINEERING

The Fu Foundation School of Engineering and Applied Science

MilkShaker™

**An automated, physical therapist-backed wearable
to prevent and treat mastitis in breastfeeding women.**

By Women, For Women.



20.8 Million

breastfeeding mothers globally struggle
with **clogged milk ducts** and **mastitis**

Never heard of it? *Neither had we.*

Postpartum Changes Can Develop into Painful Mastitis.

Postpartum changes (posture, mobility, injuries)



Lymph fluid collects in milk
ducts, restricting flow



affects 1 in 5
breastfeeding women

Clogged Milk Ducts



Infection or Inflammation



affects 1 in 10
breastfeeding women

Mastitis

(high fevers, excessive pain, etc.)

No product exists that provides complete, effective prevention and treatment.



**No Automation,
Requires Training**

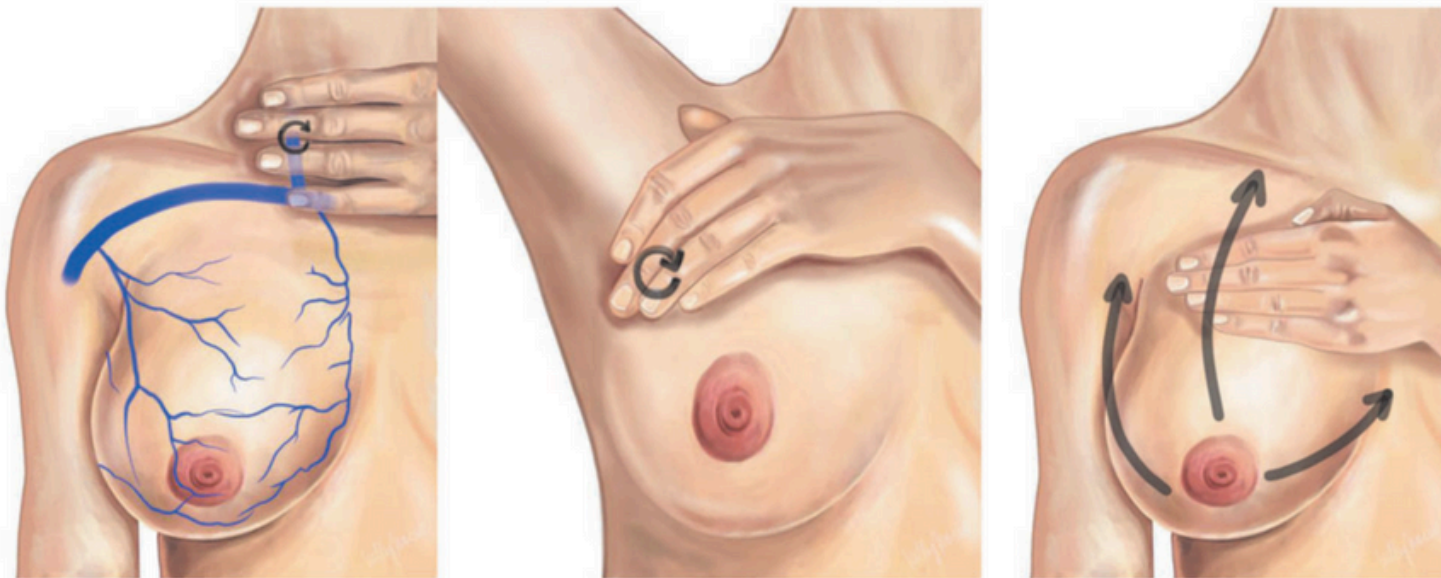


**Not Proactive
or Targeted**



**No Mobility,
Not Breast-Specific**

Lymphatic Drainage



Certified Manual Lymphatic Drainage (MLD) massages are the **only proven treatment**, but remain **inaccessible** and **expensive**.

How can we automate this process?

MilkShaker™

*both preventative and a treatment device
targeting the root cause of mastitis*

- ✓ Accessible
- ✓ Affordable
- ✓ Automated
- ✓ Proactive
- ✓ Breast-Targeted



Automated Professional Massage

*the first physical therapist-backed
breast massager for mastitis relief*

Direct “Pushing” Motion

*to painlessly guide fluid towards
underarm lymph node*

MilkShakerTM

Washable, Removable Technology

*removable mechanical inserts for machine
washing and part replacement*

Portable & Hands-Free

*wireless, rechargeable battery and compact
design offer mobility for busy mothers*



Dr. Kristin Myers

Columbia Professor, Mother



Caitha Barr

*OTR/L, CLT, Occupational Therapist,
Certified in Mastitis MLD Massage*



Dr. Reena Parikh

MD, Obstetrics and Gynecology

A product driven by professional and user input



Dr. Harry West

Columbia Engineering Professor



Fem First Health

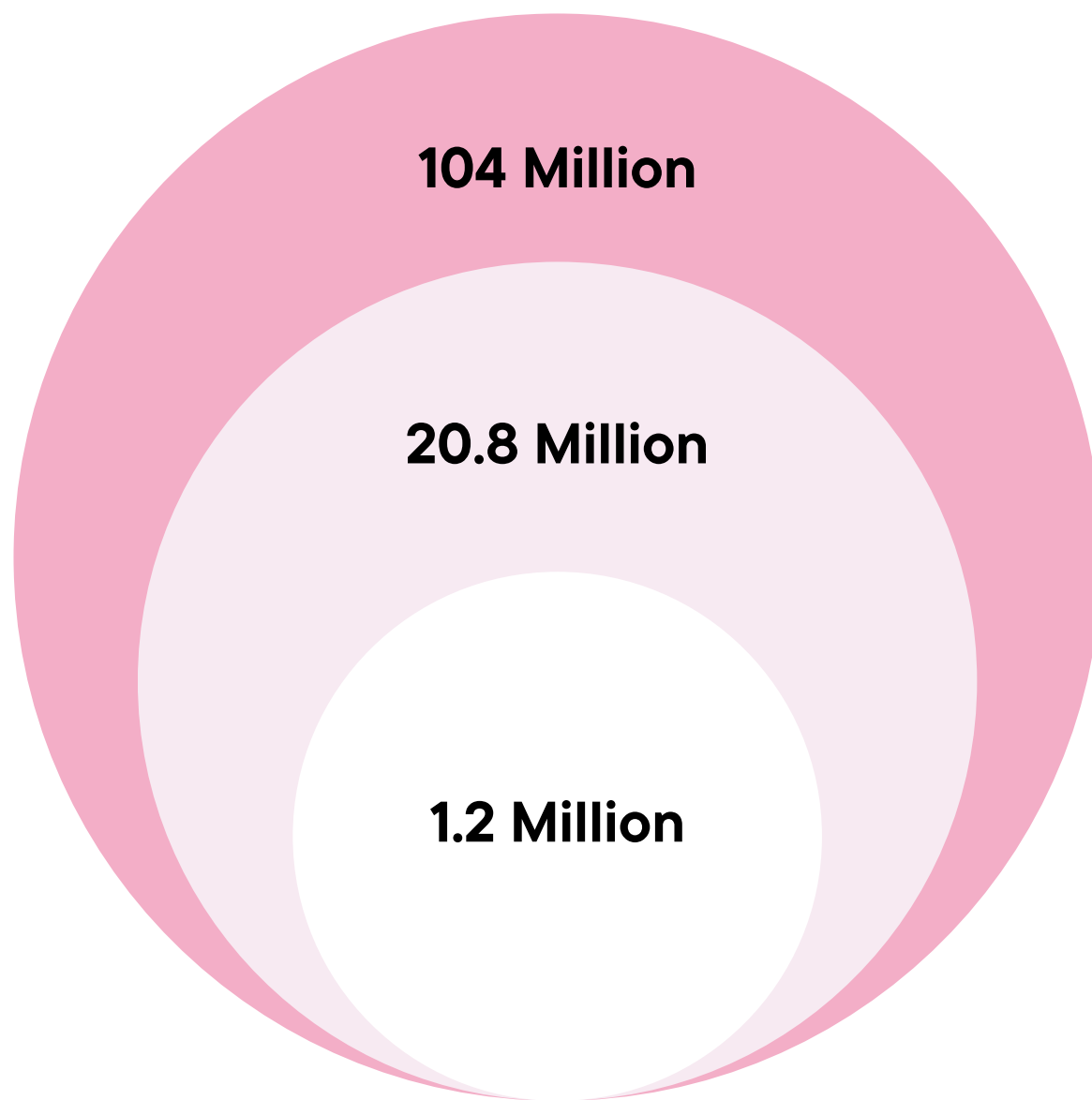


Dr. Yevgeniy Yesilevskiy

Columbia Engineering Professor

... with a market to match.

First year: ~1% penetration → \$2.4M/year
Five years: 10% penetration → \$24M/year
(At projected \$200 price point, equivalent to a standard pump.)



of Potential Users



Total Addressable Market (TAM):

Potential Annual Revenue: 20.8 Billion USD

- Worldwide births: ~130M per year
- 80% attempt breastfeeding and face mastitis risk → 104M



Serviceable Addressable Market (SAM):

Potential Annual Revenue: 4.2 Billion USD

- 20% experience clogged duct, likely to seek treatment → 20.8M



Serviceable Obtainable Market (SOM):

Potential Annual Revenue: 240 Million USD

- Focusing on US/Europe Market → 1.2M
- (Likely to expand to other markets easily due to B2C approach)

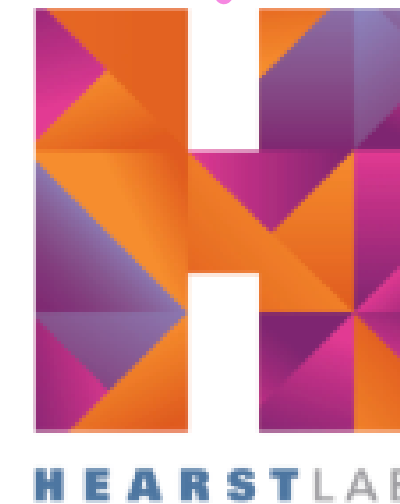
Accomplishments to Date

- * Expert Input, Market Research, Initial Brainstorming
- * Mechanical Design and Concept
- * Physical Prototyping and User Input
- * Working Prototype



**Millard Chan '99
Technology Challenge
2nd Place**

\$15K



**Helen Gurley Brown
(HGB) Foundation
3rd Place**

Next Steps

- * Provisional Patent Application
- * Integrated Wearable
- * Beta-Testing with FemFirst Patients

Launch to Market



FSA/HSA Eligibility



Insurance Coverage under
ACA Breastfeeding Clause

Now is the time to get involved.



Kavita Parikh
CEO, Co-Founder



Valentina Marini Fichera
CTO, Co-Founder

Our MilkShaker mission is to bridge the women's health gap, one device at a time.
We are committed to delivering affordable, effective treatments designed by women for women.



Pricilla Dua
Co-Founder, Engineer



Hope Hersom
Co-Founder, Engineer



Elise Yang
Co-Founder, Engineer